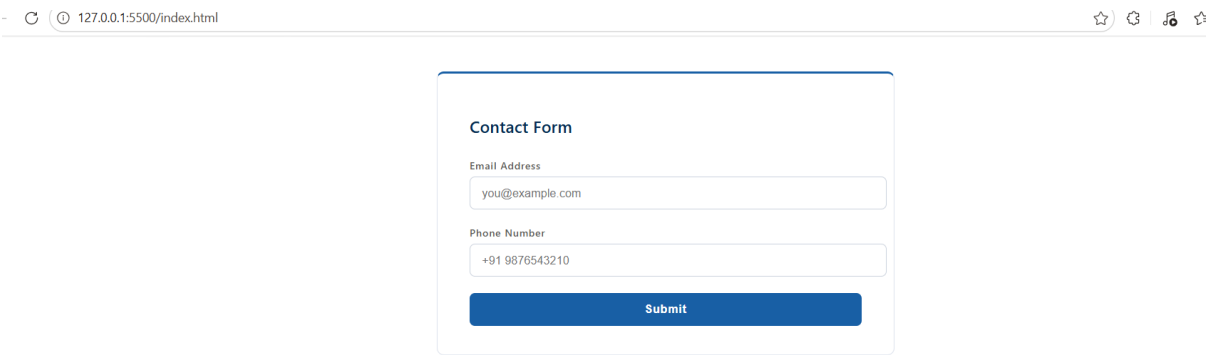
Experiment 9: Form Validation

Roll No: 240701363 Name : Nishani B

1.Initial Form UI (Before Validation)



Description

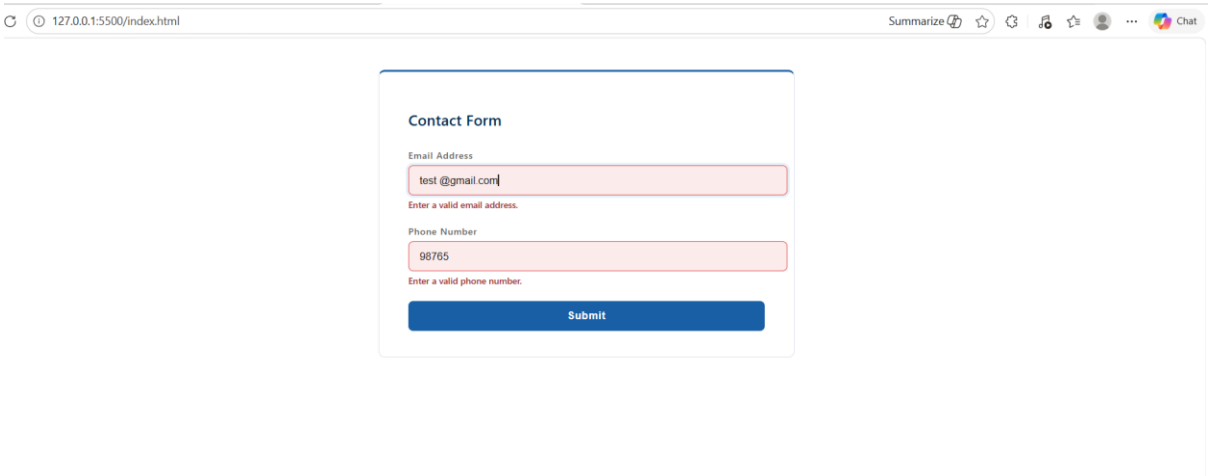
A basic contact form is created using HTML and CSS. It contains:

- Email input field
- Phone number input field
- Submit button

Output Explanation

- The form is displayed in the center of the screen.
- No validation is triggered yet.
- Input fields are empty and neutral (default styling).
- No error messages are visible.

2.Invalid Input (Error State)



Description

When the user enters incorrect input:

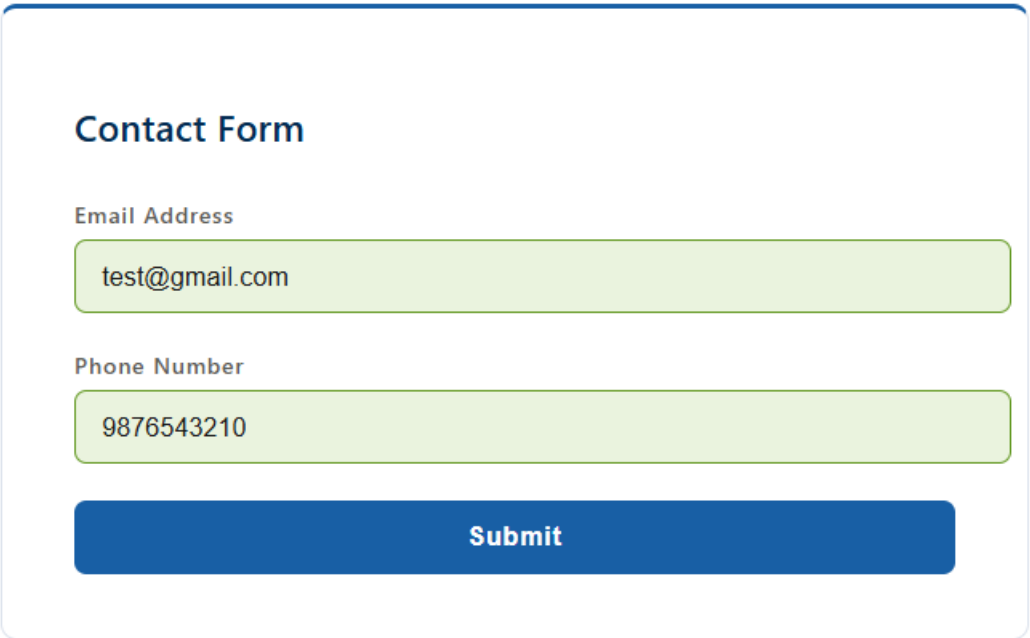
- Invalid email (e.g., abc)
- Invalid phone number (e.g., 123)

JavaScript validation is triggered.

Output Explanation

- The input field border turns red
- Background changes to light red
- Error message appears below input:
    - “Enter a valid email address”
    - “Enter a valid phone number”
- Form submission is prevented

3.Valid Input



Description

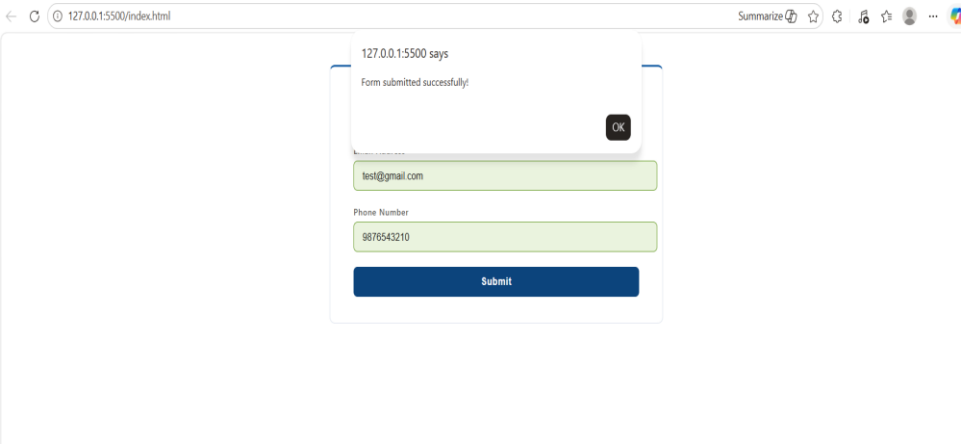
When correct data is entered:

- Email - test@gmail.com
- Phone - 9876543210

Output Explanation

- Input fields turn green
- Background becomes light green
- Error messages disappear
- Indicates valid data entry

4.Successful Form Submission



Description

When all fields are valid and user clicks Submit.

Output Explanation

- Form is successfully validated
- A popup appears:
  - “Form submitted successfully!”
- No errors are shown
- Demonstrates complete validation flow

Code:

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">
```

```
  <title>Form Validation</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">

    <h2>Contact Form</h2>

    <form id="myForm" novalidate>

      <!-- Email -->
      <div class="form-group">
        <label for="email">Email Address</label>
        <input type="email" id="email" placeholder="you@example.com">
        <span class="error" id="emailErr"></span>
      </div>

      <!-- Phone -->
      <div class="form-group">
        <label for="phone">Phone Number</label>
        <input type="tel" id="phone" placeholder="+91 9876543210">
        <span class="error" id="phoneErr"></span>
      </div>

      <button type="submit">Submit</button>

    </form>

  </div>
```

```
  <script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.9.0/validator.min.js"></script>

  <!-- Your JS -->
  <script src="script.js"></script>

</body>
</html>
```

Script.js

```
const emailInput = document.getElementById("email");
const phoneInput = document.getElementById("phone");
const emailErr   = document.getElementById("emailErr");
const phoneErr   = document.getElementById("phoneErr");
function setError(input, span, msg) {
  input.classList.remove("success");
  input.classList.add("error");
  span.textContent = msg;
}
function setSuccess(input, span) {
  input.classList.remove("error");
  input.classList.add("success");
  span.textContent = "";
}

// Email validation
function validateEmail() {
  const val = emailInput.value.trim();
```

```
  if (!val) {
    setError(emailInput, emailErr, "Email is required.");
    return false;
  }

  if (!validator.isEmail(val)) {
    setError(emailInput, emailErr, "Enter a valid email address.");
    return false;
  }

  setSuccess(emailInput, emailErr);
  return true;
}

// Phone validation
function validatePhone() {
  const val = phoneInput.value.trim();

  if (!val) {
    setError(phoneInput, phoneErr, "Phone number is required.");
    return false;
  }

  if (!validator.isMobilePhone(val, "any", { strictMode: false })) {
    setError(phoneInput, phoneErr, "Enter a valid phone number.");
    return false;
  }

  setSuccess(phoneInput, phoneErr);
  return true;
```

```
}
emailInput.addEventListener("input", validateEmail);
phoneInput.addEventListener("input", validatePhone);
document.getElementById("myForm").addEventListener("submit", function (e) {
  e.preventDefault();

  const isValid = validateEmail() && validatePhone();

  if (isValid) {
    alert("Form submitted successfully!");
  }
});
```

Style.css

```
.container {
  max-width: 440px;
  margin: 40px auto;
  padding: 32px 36px;
  border: 1px solid #dde3ed;
  border-top: 3px solid #185FA5;
  border-radius: 8px;
  font-family: 'Segoe UI', sans-serif;
}

h2 {
  font-size: 18px;
  font-weight: 600;
  color: #042C53;
  margin-bottom: 24px;
  letter-spacing: .02em;
```

```
}

.form-group {
  margin-bottom: 18px;
}

label {
  display: block;
  font-size: 11px;
  font-weight: 600;
  letter-spacing: .07em;
  color: #5F5E5A;
  margin-bottom: 6px;
}

input[type="email"],
input[type="tel"] {
  width: 100%;
  padding: 10px 13px;
  border: 1px solid #cdd4df;
  border-radius: 6px;
  font-size: 13px;
  color: #1a1a1a;
  outline: none;
  transition: border-color .2s, box-shadow .2s;
}
input:focus {
  border-color: #185FA5;
  box-shadow: 0 0 0 3px #E6F1FB;
}
```

```
/* Error state */
input.error {
  border-color: #E24B4A;
  background: #FCEBEB;
}

span.error {
  display: block;
  font-size: 11px;
  font-weight: 600;
  color: #A32D2D;
  margin-top: 4px;
}
input.success {
  border-color: #639922;
  background: #EAF3DE;
}

button[type="submit"] {
  width: 100%;
  padding: 11px;
  background: #185FA5;
  color: #fff;
  border: none;
  border-radius: 6px;
  font-size: 13px;
  font-weight: 600;
  letter-spacing: .04em;
  cursor: pointer;
```

```
  transition: background .15s;
}

button:hover {
  background: #0C447C;
}
```